Apex Retail Intelligence

E2E Video Analytics & Store Optimization API

Purplle Tech Challenge 2026

Architecture Overview

1. **Computer Vision (Edge)**
    - **YOLOv8 & ByteTrack:** High-speed real-time detection & temporal tracking of individuals.
    - **Coordinate Mapping:** Translates pixel coordinates into store zones (Entry, Zone Visit, Billing).
2. **Intelligence API (Cloud/Server)**
    - **FastAPI Backend:** Handles heavy ingestion loads safely.
    - **SQLite Database:** Zero-configuration, scalable persistence layer.

Addressing Edge Cases

- **Staff Exclusion:** Employs explicit `is_staff` schema tagging. Filtered out of all metric pipelines.
- **Re-Entry Handling:** Uses mathematical `Set()` aggregation by `visitor_id` to prevent conversion funnel inflation.
- **Queue Buildup:** Continuously calculates Queue Depth. Pushes dynamic `BILLING_QUEUE_SPIKE` alerts to operations.
- **Occlusion Resilience:** Pushes detection `confidence` scores natively to the database for post-filtering.

The Conversion Funnel

The system actively maps POS transactions (`pos_transactions.csv`) against the raw video-extracted timeline to provide a literal 4-stage checkout funnel:

1. **Entry** (Crossed Threshold)
2. **Zone Visit** (Engaged with layout)
3. **Billing Queue** (Intent to buy)
4. **Purchase** (Confirmed POS match)

Deployment & Scalability

- **100% Dockerized:** Instantly runs on any system using `docker-compose up`.
- **Stateless Intelligence:** The FastAPI server maintains zero state, meaning it can be load-balanced horizontally across multiple store locations.
- **Extensible:** The API exposes clean JSON endpoints for custom dashboards and alerts.